



2013
PRELIMINARY
EXAMINATION

Student Number: _____

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen. Black pen is preferred.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 6–9.

Total Marks – 65

Section I

Pages 2–3

5 marks

- Attempt Questions 1–5
- Allow about 10 minutes for this section

Section II

Pages 4–7

60 marks

- Attempt Questions 6–9
- Allow about 1 hours 50 minutes for this section

Section I

5 marks

Attempt Questions 1–5

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

1 What is the value of $\sqrt{9.87^2 - 1.23^2}$, correct to 3 significant figures?

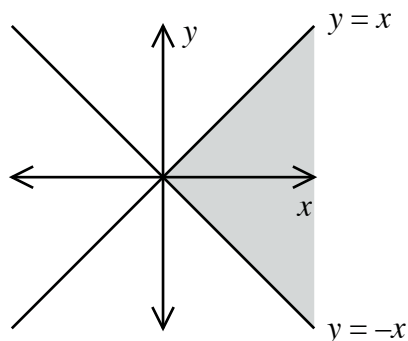
(A) 2.94

(B) 8.64

(C) 9.79

(D) 9.95

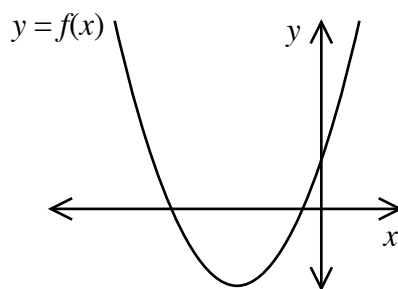
2



Which two inequalities best describe the shaded region shown in the graph above?

(A) $y \geq x$ and $y \geq -x$ (B) $y \geq x$ and $y \leq -x$ (C) $y \leq x$ and $y \geq -x$ (D) $y \leq x$ and $y \leq -x$

3 The diagram below shows the graph of a quadratic function.



Which of the following statements is true?

(A) $a > 0$ and $\Delta > 0$ (B) $a > 0$ and $\Delta < 0$ (C) $a < 0$ and $\Delta > 0$ (D) $a < 0$ and $\Delta < 0$

4 What is the angle of inclination of the line $3x - 5y - 7 = 0$ with the positive direction of the x -axis?

(A) $30^\circ 58'$

(B) $59^\circ 2'$

(C) $120^\circ 58'$

(D) $149^\circ 2'$

5 Which of these expressions is equal to $\tan(180 - \theta)^\circ \times \tan(90 - \theta)^\circ$?

(A) 1

(B) -1

(C) $\tan^2\theta$

(D) $-\tan^2\theta$

Section II

60 marks

Attempt Questions 6–9

Allow about 1 hours 50 minutes for this section

Start each question on a new page. Extra writing pages are available.

All necessary working should be shown in every question.

Question 6

(15 marks)

Start a new page

(a) Solve the following:

(i) $\frac{x}{3} - \frac{2x-1}{4} = 1$ **2**

(ii) $|2x-9| = 3x-1$ **3**

(b) Express $\frac{10-\sqrt{3}}{10+\sqrt{3}}$ as a fraction with a rational denominator in simplest form. **2**

(c) Simplify: $\frac{x^2-81}{x^3+27} \times \frac{x+3}{x-9}$ **2**

(d) If $\sin A = \frac{4}{5}$ and $\cos A < 0$, find the exact value of $\tan A$. **2**

(e) α and β are the roots of the equation $2x^2 - 3x + 5 = 0$. Find the value of:

(i) $\alpha + \beta$ **1**

(ii) $\alpha\beta$ **1**

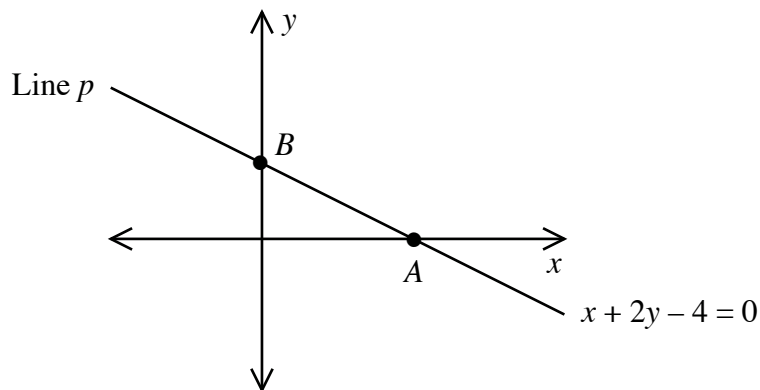
(iii) $\alpha^2 + \beta^2$ **2**

Question 7

(15 marks)

[Start a new page](#)

- (a) In the diagram below, the line p has equation $x + 2y - 4 = 0$, and cuts the x -axis and y -axis at A and B respectively.



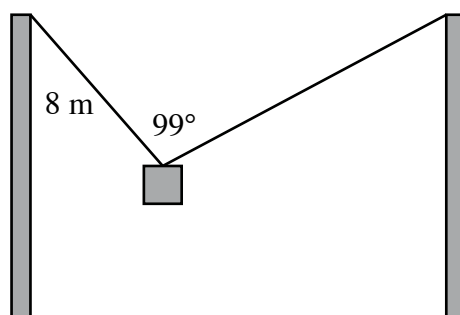
- (i) Copy the diagram onto your writing paper.
- (ii) Find the coordinates of A and B . **2**
- (iii) Find the gradient of AB . **1**
- (iv) Show that the equation of the line q , perpendicular to the line AB and passing through the point A is $2x - y - 8 = 0$. **3**
- (v) Find the area of the triangle enclosed by line q and the coordinate axes. **2**
- (b) Solve the following:
- (i) $3x^2 - 5x - 28 < 0$ **2**
- (ii) $x^6 - 28x^3 + 27 = 0$ **3**
- (c) State the domain and range of the function $y = \sqrt{2x - 6}$. **2**

Question 8

(15 marks)

[Start a new page](#)

- (a) For what value(s) of k will the equation $x^2 + (k + 3)x + 4k = 0$ have:
- (i) roots which are equal? 2
 - (ii) roots which are reciprocals of each other? 2
- (b) Find the values of A , B and C if $x^2 + x + 1 \equiv A(x - 2)^2 + B(x - 2) + C$. 3
- (c) A rope 20 metres long is tied to the top of two posts of equal height fixed into level ground. A weight is attached to the rope so that it is 8 metres from one end of the rope. The angle between the two parts of the rope is 99° , as shown in the diagram below. 3



Find the distance between the two posts. Give your answer to correct to 1 decimal place.

- (d) A function $f(x)$ is defined as:
- $$f(x) = \begin{cases} 5x - 4 & \text{for } x \geq 0 \\ px + q & \text{for } x < 0 \end{cases}$$
- (i) Evaluate $f(2)$. 1
 - (ii) Find an expression for $f(3) + f(-2)$. Give your answer in simplest form. 2
 - (iii) Find the values of p and q so that $f(x)$ is an even function. 2

Question 9

(15 marks)

[Start a new page](#)

- (a) Show that $\frac{x+2}{x+1} = 1 + \frac{1}{x+1}$. **2**
- (b) Solve the equation $2 \sin 2x - 1 = 0$ in the domain $0^\circ \leq x \leq 360^\circ$. **3**
- (c) (i) Factorise $x^2 + 3x + 2$. **1**
- (ii) Prove that $\frac{\sin^2 x + 3 \sin x + 2}{\cos^2 x} = \frac{2 + \sin x}{1 - \sin x}$. **3**
- (d) (i) Fully factorise the expression $2x^2 + 4x - 6$. **1**
- (ii) Find the minimum value of the function $f(x) = 2x^2 + 4x - 6$. **1**
- (iii) Hence, or otherwise, draw the graph of the function $y = |2x^2 + 4x - 6|$. Use at least half a page for your graph. Ensure that you label the key features of the graph. **2**
- (iv) On the same diagram, draw the graph of the function $y = x + 2$. **1**
- (v) Hence state the **number** of solutions for the equation $x + 2 = |2x^2 + 4x - 6|$. **1**

End of paper

2 UNIT MATHEMATICS
2013 PRELIMINARY EXAMINATION

SECTION I

$$\begin{aligned}
 1 \quad \sqrt{9.87^2 - 1.23^2} &= \sqrt{9.87^2 - 1.23^2} & 1. \quad \mathbf{C} \\
 &= \sqrt{95.904} \\
 &= 9.793058766 \\
 &\approx 9.79
 \end{aligned}$$

$\therefore$ Answer: **C**

$$\begin{aligned}
 2 \quad \text{Choosing } (1,0) \text{ as a test point:} & & 2. \quad \mathbf{C} \\
 \text{For } y = x, \text{ we have } 0 < 1, \text{ so the region is defined as } y \leq x. \\
 \text{For } y = -x, \text{ we have } 0 > -1, \text{ so the region is defined as } y \geq -x. \\
 \therefore \text{ Answer: } \mathbf{C}
 \end{aligned}$$

$$\begin{aligned}
 3 \quad \text{Parabola is concave up, so } a > 0. & & 3. \quad \mathbf{A} \\
 \text{Parabola has two zeros, so } \Delta > 0. \\
 \therefore \text{ Answer: } \mathbf{A}
 \end{aligned}$$

$$\begin{aligned}
 4 & & 4. \quad \mathbf{A} \\
 \text{Gradient} &= \frac{-a}{b} \\
 &= \frac{-3}{-5} \\
 &= \frac{3}{5} \\
 \tan \theta &= m \\
 &= \frac{3}{5} \\
 \theta &= 30.96375653 \\
 &= 30^\circ 58' \\
 \therefore \text{ Answer: } \mathbf{A}
 \end{aligned}$$

$$\begin{aligned}
 5 \quad \tan (180 - \theta) \times \tan (90 - \theta) &= -\tan \theta \times \cot \theta & 5. \quad \mathbf{B} \\
 &= -\frac{\sin \theta}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} \\
 &= -1
 \end{aligned}$$

$\therefore$ Answer: **B**

SECTION II

QUESTION 6

(a) (i)

$$\begin{aligned}\frac{x}{3} - \frac{2x-1}{4} &= 1 \\ \frac{4x}{12} - \frac{3(2x-1)}{12} &= 1 \\ \frac{4x-3(2x-1)}{12} &= 1 \\ \frac{4x-6x+3}{12} &= 1 \\ \frac{-2x+3}{12} &= 1 \\ -2x+3 &= 12 \\ -2x &= 9 \\ x &= -\frac{9}{2}\end{aligned}$$

(ii) When $2x - 9 \geq 0$ (or $x \geq 4\frac{1}{2}$):

$$\begin{aligned}2x - 9 &= 3x - 1 \\ 2x &= 3x + 8 \\ -x &= 8 \\ x &= -8\end{aligned}$$

But since $x \geq 4\frac{1}{2}$, there is no solution in this case.

When $2x - 9 < 0$ (or $x < 4\frac{1}{2}$):

$$\begin{aligned}-(2x - 9) &= 3x - 1 \\ -2x + 9 &= 3x - 1 \\ -2x &= 3x - 10 \\ -5x &= -10 \\ x &= 2\end{aligned}$$

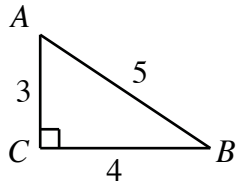
This is a valid solution.

$\therefore$ The only solution is $x = 2$.

$$\begin{aligned}\text{(b)} \quad \frac{10-\sqrt{3}}{10+\sqrt{3}} &= \frac{10-\sqrt{3}}{10+\sqrt{3}} \times \frac{10-\sqrt{3}}{10-\sqrt{3}} \\ &= \frac{(10-\sqrt{3})^2}{(10+\sqrt{3})(10-\sqrt{3})} \\ &= \frac{100-20\sqrt{3}+3}{(10)^2-(\sqrt{3})^2} \\ &= \frac{103-20\sqrt{3}}{100-3} \\ &= \frac{103-20\sqrt{3}}{97}\end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad \frac{x^2 - 81}{x^3 + 27} \times \frac{x-9}{x+3} &= \frac{(x+9)(x-9)}{(x+3)(x^2 - 3x + 9)} \times \frac{x+3}{x-9} \\
 &= \frac{(x+9)\cancel{(x-9)}}{\cancel{(x+3)}(x^2 - 3x + 9)} \times \frac{\cancel{x+3}}{\cancel{x-9}} \\
 &= \frac{x+9}{x^2 - 3x + 9}
 \end{aligned}$$

(d)



Since $\sin A > 0$ and $\cos A < 0$, A lies in the 2nd quadrant.

$$\therefore \tan A < 0.$$

$$\therefore \tan A = -\frac{4}{3}$$

$$\text{(e)} \quad 2x^2 - 3x + 5 = 0$$

$$\therefore a = 2, b = -3, c = 5$$

(i)

$$\begin{aligned}
 \alpha + \beta &= \frac{-b}{a} \\
 &= \frac{-(-3)}{2} \\
 &= \frac{3}{2}
 \end{aligned}$$

(ii)

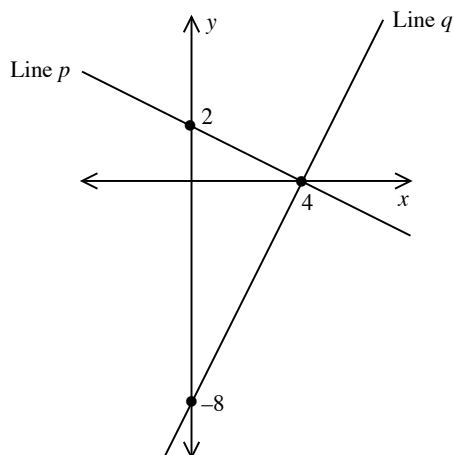
$$\begin{aligned}
 \alpha\beta &= \frac{c}{a} \\
 &= \frac{5}{2}
 \end{aligned}$$

(iii)

$$\begin{aligned}
 \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\
 &= \left(\frac{3}{2}\right)^2 - 2\left(\frac{5}{2}\right) \\
 &= \frac{9}{4} - 5 \\
 &= -\frac{11}{4}
 \end{aligned}$$

QUESTION 7

(a) (i)



(ii) When $y = 0$:

$$x + 2(0) - 4 = 0$$

$$x - 4 = 0$$

$$x = 4$$

$$\therefore A \equiv (4, 0)$$

When $x = 0$:

$$(0) + 2y - 4 = 0$$

$$2y - 4 = 0$$

$$2y = 4$$

$$y = 2$$

$$\therefore B \equiv (0, 2)$$

(iii) Gradient of $AB = \frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{2 - 0}{0 - 4}$$

$$= -\frac{1}{2}$$

or

$$\text{Gradient of } AB = \frac{-a}{b}$$

$$= \frac{-1}{2}$$

$$= -\frac{1}{2}$$

(iv) Since line q is perpendicular to line p the gradient of line q is 2.

Equation of q :

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 2(x - 4)$$

$$y = 2x - 8$$

$$2x - y - 8 = 0$$

or

Equation of line p is $x + 2y - 4 = 0$.

$\therefore$ Equation of line q is $2x - y + k = 0$.

Since line q passes through $(4, 0)$:

$$\begin{aligned}
2(4) - (0) + k &= 0 \\
8 - 0 + k &= 0 \\
8 + k &= 0 \\
k &= -8
\end{aligned}$$

$\therefore$ Equation of line q is $2x - y - 8 = 0$.

(v) When $x = 0$:

$$\begin{aligned}
2(0) - y - 8 &= 0 \\
-y - 8 &= 0 \\
y &= -8
\end{aligned}$$

$\therefore$ Line q cuts the y -axis at $(0, -8)$.

$$\begin{aligned}
\text{Area} &= \frac{1}{2}bh \\
&= \frac{1}{2} \times 4 \times 8 \\
&= 16 \text{ square units}
\end{aligned}$$

(b) (i) Consider $3x^2 - 5x - 28 = 0$

$$(3x + 7)(x - 4) = 0$$

$$\therefore x = -2\frac{1}{3} \text{ or } 4$$

$\therefore$ The solution to the inequality $3x^2 - 5x - 28 < 0$ is $-2\frac{1}{3} < x < 4$.

(ii) $x^6 - 28x^3 + 27 = 0$

$$(x^3)^2 - 28x^3 + 27 = 0$$

$$\text{Let } t = x^3$$

$$t^2 - 28t + 27 = 0$$

$$(t - 1)(t - 27) = 0$$

$$\therefore t = 1 \text{ or } 27$$

$$\therefore x^3 = 1 \text{ or } 27$$

$$\therefore x = 1 \text{ or } 3$$

(c) For the domain:

$$2x - 6 \geq 0$$

$$2x \geq 6$$

$$x \geq 3$$

For the range:

$$y \geq 0$$

QUESTION 8

- (a) (i) For equal roots, $\Delta = 0$. Now:

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= (k+3)^2 - 4(1)(4k) \\ &= k^2 + 6k + 9 - 16k \\ &= k^2 - 10k + 9\end{aligned}$$

Therefore we need:

$$\begin{aligned}k^2 - 10k + 9 &= 0 \\ (k-1)(k-9) &= 0 \\ k &= 1 \text{ or } 9\end{aligned}$$

- (ii) For roots as reciprocals, their product of the roots is 1. Now:

$$\begin{aligned}\text{Product of roots} &= \frac{c}{a} \\ &= \frac{4k}{1} \\ &= 4k\end{aligned}$$

Therefore we need:

$$\begin{aligned}4k &= 1 \\ k &= \frac{1}{4}\end{aligned}$$

$$\begin{aligned}\text{(b) } x^2 + x + 1 &\equiv A(x-2)^2 + B(x-2) + C \\ &\equiv A(x^2 - 4x + 4) + B(x-2) + C \\ &\equiv Ax^2 - 4Ax + 4A + Bx - 2B + C \\ &\equiv Ax^2 - 4Ax + Bx + 4A - 2B + C \\ &\equiv Ax^2 + (-4A + B)x + (4A - 2B + C)\end{aligned}$$

Therefore:

$$A = 1$$

and:

$$-4A + B = 1$$

$$-4(1) + B = 1$$

$$-4 + B = 1$$

$$B = 5$$

and:

$$4A - 2B + C = 1$$

$$4(1) - 2(5) + C = 1$$

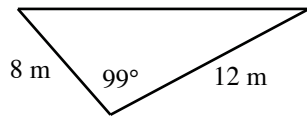
$$4 - 10 + C = 1$$

$$-6 + C = 1$$

$$C = 7$$

$$\therefore A = 1, B = 5, C = 7$$

- (c) Since the rope is 20 m long, the two portions of rope must be 8 m and 12 m.



Let d be the distance between the posts.

Using the cosine rule:

$$\begin{aligned} d^2 &= 8^2 + 12^2 - 2 \times 8 \times 12 \times \cos 99^\circ \\ &= 238.0354173 \\ d &= 15.42839646 \\ &= 15.4 \end{aligned}$$

$\therefore$ The posts are 15.4 m apart.

- (d) (i) $f(2) = 5(2) - 4$
 $= 6$
- (ii) $f(3) + f(-2) = [5(3) - 4] + [p(-2) + q]$
 $= 15 - 4 - 2p + q$
 $= 11 - 2p + q$
- (iii) An even function is symmetrical about the y axis.
 The line $y = px + q$ has a gradient which is the same magnitude but opposite sign compared to $y = 5x - 4$, and touches the y axis at the same point.
 $\therefore p = -5, q = -4$

QUESTION 9

(a)

$$\begin{aligned} \text{RHS} &= 1 + \frac{1}{x+1} \\ &= \frac{x+1}{x+1} + \frac{1}{x+1} \\ &= \frac{x+1+1}{x+1} \\ &= \frac{x+2}{x+1} \\ &= \text{LHS} \end{aligned}$$

or

$$\begin{aligned} \text{LHS} &= \frac{x+2}{x+1} \\ &= \frac{x+1+1}{x+1} \\ &= \frac{x+1}{x+1} + \frac{1}{x+1} \\ &= 1 + \frac{1}{x+1} \\ &= \text{RHS} \end{aligned}$$

(b)

$$2 \sin 2x - 1 = 0$$

$$2 \sin 2x = 1$$

$$\sin 2x = \frac{1}{2}$$

$$2x = 30^\circ, 150^\circ, 390^\circ \text{ or } 510^\circ$$

$$x = 15^\circ, 75^\circ, 195^\circ \text{ or } 255^\circ$$

(c) (i) $x^2 + 3x + 2 = (x + 1)(x + 2)$

(ii) $\text{LHS} = \frac{\sin^2 x + 3 \sin x + 2}{\cos^2 x}$

$$= \frac{(\sin x + 1)(\sin x + 2)}{1 - \sin^2 x}$$

$$= \frac{(\cancel{\sin x + 1})(\sin x + 2)}{(\cancel{1 + \sin x})(1 - \sin x)}$$

$$= \frac{2 + \sin x}{1 - \sin x}$$

$$= \text{RHS}$$

(d) (i) $2x^2 + 4x - 6 = 2(x^2 + 2x - 3)$
 $= 2(x + 3)(x - 1)$

(ii) Zeros of function are -3 and 1 .

$\therefore$ Minimum value exists when $x = -1$.

$$\text{Minimum value} = 2(-1)^2 + 4(-1) - 6$$

$$= 2 - 4 - 6$$

$$= -8$$

or

$$\text{For minimum value, } x = \frac{-b}{2a}$$

$$= \frac{-4}{2(2)}$$

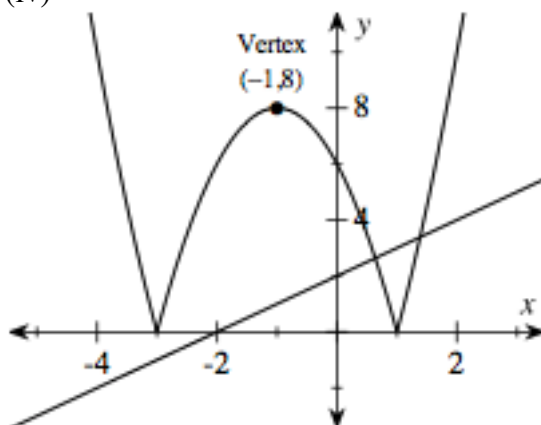
$$= -1$$

$$\text{Minimum value} = 2(-1)^2 + 4(-1) - 6$$

$$= 2 - 4 - 6$$

$$= -8$$

(iii), (iv)



(iv) The equation $x + 2 = |2x^2 + 4x - 6|$ has 2 solutions.